

EARTHQUAKE-INDUCED LANDSLIDING IN NEW ZEALAND AND POTENTIAL FOR LANDSLIDES DURING EARTHQUAKES IN ADELAIDE, SOUTH AUSTRALIA

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ABSTRACT

A study of landsliding caused by historical earthquakes in New Zealand was completed at the end of 1997. That study showed the minimum magnitude for earthquake-induced landsliding (EIL) in N.Z. to be M 5, with significant landsliding at M 6 or greater. The minimum MM intensity for landsliding is MM6 and the most common intensities for landslides are MM7–8. The threshold for liquefaction is MM7 for sand boils and MM8 for lateral spreading. Environmental criteria (landslides, liquefaction) were also defined for the MM Intensity Scale.

In this paper the EIL relationships and MM Intensity criteria are described and then applied in Adelaide. This indicates that an M 6–7 (MM8–9) earthquake in the Adelaide area could cause moderate to large rock falls and slides on high unsupported cuts, river banks, terrace edges and coastal cliffs. Liquefaction damage could also occur in areas of saturated sandy alluvium and estuarine deposits. The historical seismic record, however, suggests that the probability of an M 6 or M 7 earthquake in Adelaide is relatively low; hence the potential risk from earthquake-induced landslides and liquefaction is also likely to be low.

1 INTRODUCTION

Landslides have occurred during many historical earthquakes in New Zealand. Since 1840, at least 22 earthquakes have resulted in widespread and damaging landsliding (Figure 1). The building damage caused by earthquake-induced landsliding (EIL) has been second only to damage caused by strong shaking. Buildings have been destroyed, and many roads have been damaged and closed by landslides and rock falls. In the last 80 years at least seventeen people have been killed by earthquake-triggered landslides in New Zealand, with 16 of those deaths occurring during the 1929 Murchison earthquake.

A comprehensive study of landsliding caused by 22 New Zealand earthquakes (Figure 1) was completed in 1997 (Hancox *et al.*, 1997). The main aims of the 1997 study were to: (a) determine the nature and extent of landsliding and other ground damage (sand boils, subsidence and lateral spreading due to soil liquefaction) caused by historical earthquakes; (b) establish relationships between landslide distribution and earthquake magnitude (M), epicentre, isoseismals, faulting, geology and topography; and (c) establish improved environmental response criteria and ground classes for assigning MM (Modified Mercalli) intensities and seismic hazard assessments in New Zealand.

Maps of landslides caused by the 22 earthquakes studied were prepared using information from old geological reports, newspapers, books, papers, photographs and maps, supplemented by studies of aerial photos, topographic maps, and data in the GNS Science (GNS) database of large landslides. Locations of landslides attributed to each earthquake were incorporated in a geographic information system (GIS) and plotted on maps, along with earthquake data (epicentre, MM isoseismals, aftershocks) and any surface faulting that occurred.

A paper summarising the 1997 study (Hancox *et al.*, 1997) and related research was published in 2002 (Hancox *et al.*, 2002). New data presented in 2002 included a detailed study of landslides triggered by the M_w 7.8 Murchison 1929 earthquake, a landslide size/frequency distribution for earthquakes over the last 150 years, and an EIL Opportunity and hazard model for New Zealand. The 1929 Murchison earthquake is the most significant historical EIL event in New Zealand. As an example, a landslide map for that earthquake is shown in Figure 2 and the Matakitaiki Landslide ($\sim 18 \text{ M m}^3$), one of the very large fatal dip-slope failures triggered by the earthquake, is illustrated in Figure 3.

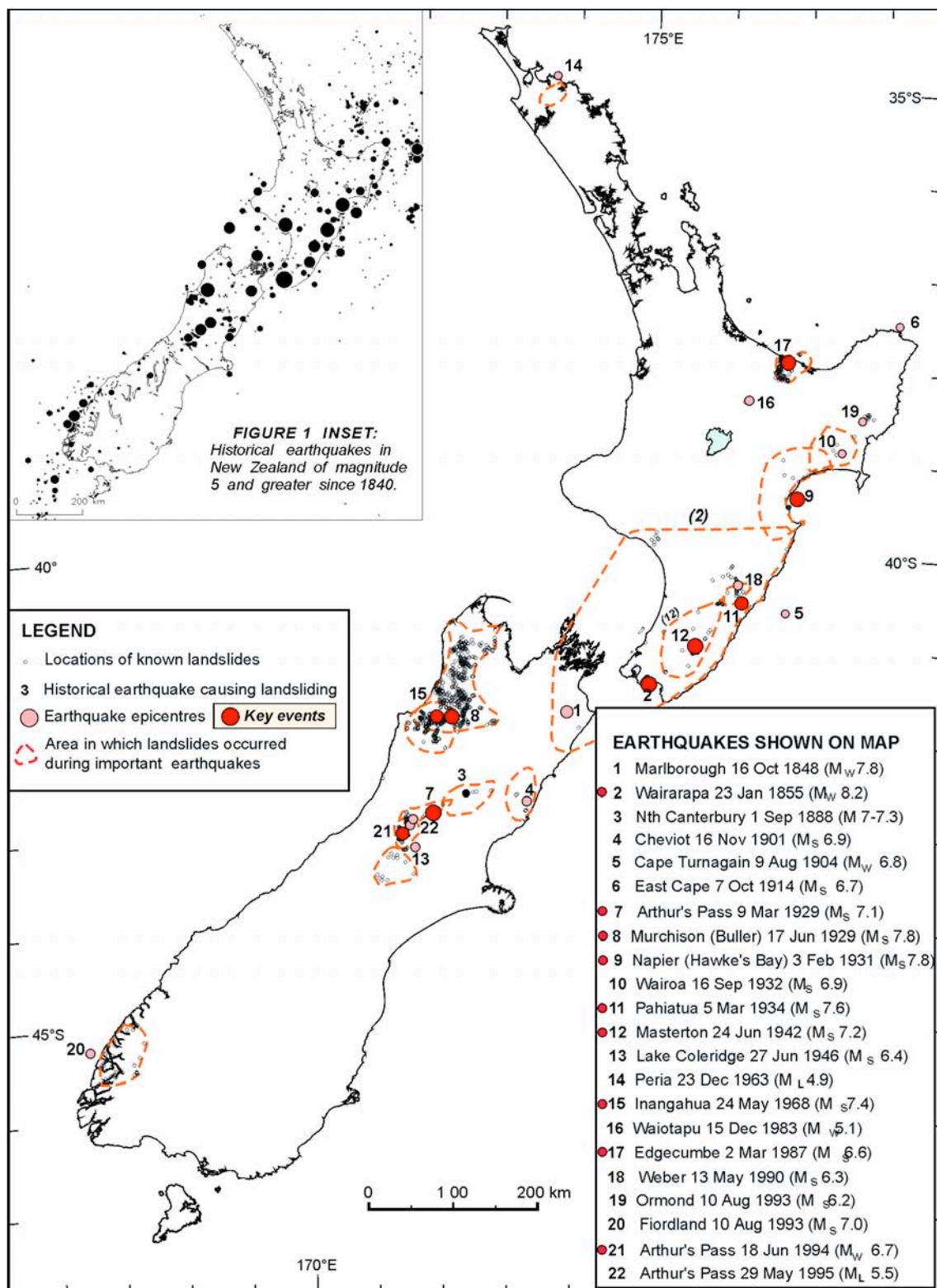


Figure 1: Map showing location of historical earthquakes that have caused substantial landsliding in New Zealand. The locations of the main landslides are indicated by small dots, and dashed lines show the landslide affected areas. Nine large earthquakes were identified as 'key events'. All of the earthquakes occurred in areas of highest seismic activity in New Zealand since 1840 (see inset map).

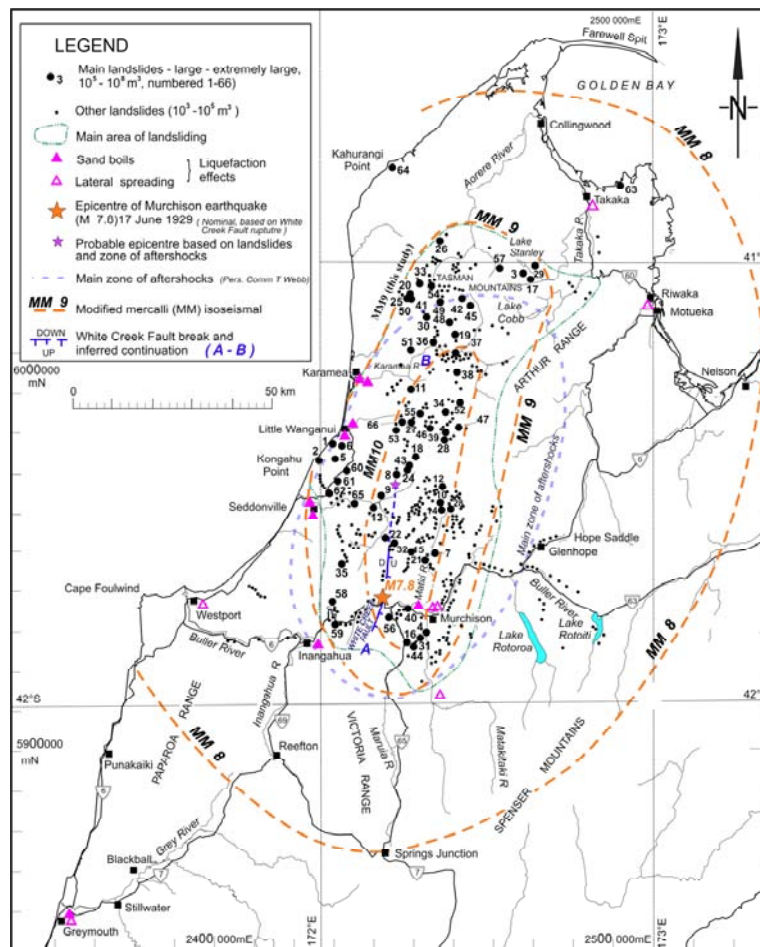


Figure 2: Map showing the locations of the main landslides and liquefaction sites resulting from by the 1929 Murchison M_w 7.8 earthquake and isoseismals (MM9 and MM10 zones) based on landslides and liquefaction effects (after Hancox et al., 2002).



Figure 3. Recent aerial view of the Matakaitiki Landslide ($\sim 18 M m^3$, within the dashed line), which was triggered by the Murchison earthquake. Debris from this very large dip-slope rock slide/ avalanche travelled ~ 1 km across the valley floor destroying two houses and killing 4 people. The top half of one house (H) was moved ~ 100 m. The landslide formed a dam with a lake extending ~ 5 km up valley. The dam lasted ten years before being washed out in a flood. The town of Murchison (M) 5 km downstream was not affected by the landslide or the dam-break flood

The present paper presents a summary of historical earthquake-induced landsliding and liquefaction in New Zealand. The types, locations, and size of landslides triggered by earthquakes and liquefaction phenomena (sand boils and lateral spreading) are described and illustrated. New Zealand relationships between landsliding and liquefaction effects and earthquake magnitude, MM shaking intensity, geology, and topography (slope angle, height, modification) are outlined. Environmental (landslides and liquefaction) criteria for the New Zealand MM Intensity Scale are also defined, and examples of EIL susceptibility studies using these relationships in Wellington are presented. These relationships are then used to estimate the potential for future earthquake-induced landsliding and liquefaction in the Adelaide area of South Australia.

2 EARTHQUAKE-INDUCED LANDSLIDE AND LIQUEFACTION RELATIONSHIPS

Relationships have been established between landslide types, size, liquefaction effects, and earthquake magnitude, MM intensity, and the areas affected by landslides during earthquakes of different magnitude in N.Z., as shown in Figures 4 and 5.

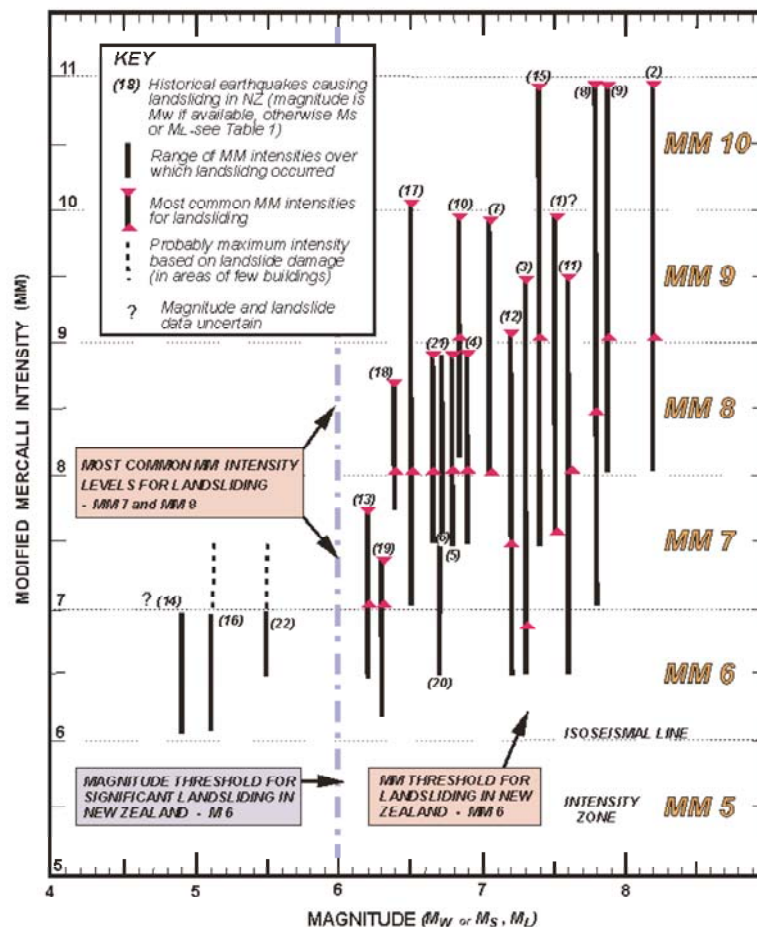


Figure 4: Relationship of earthquake magnitude and MM intensity to landslides triggered by historical earthquakes in New Zealand (from Hancox et al., 2002).

As shown in Figure 4, the minimum intensity for landsliding is MM6, while the most common intensities for significant landsliding are MM7 and MM8. Very large (10^6 m^3 or $>$) landslides occur mainly at MM9 and 10, and were used to estimate zones of MM10 shaking for the 1855 Wairarapa and 1929 Murchison earthquakes. In New Zealand, very small landslides (boulders, falls of $1\text{-}10 \text{ m}^3$) occur at M 5, but significant landsliding occurs only at M 6 or greater. Most widespread landsliding (mainly disrupted slides or falls of rock and soil) is caused by shallow earthquakes of M 6.2 to 8.2. This relationship is similar to those for overseas earthquakes (Keefer, 1984), except for slightly higher EIL thresholds in New Zealand for magnitude (M 5 cf. M 4 overseas) and MM intensity (MM 6 cf. MM 4-5 overseas).

The maximum area affected by landslides (or in which landslides occur) ranges from about 100 km^2 at M 5 to $20,000 \text{ km}^2$ at M 8.2 (Figure 5). The following mean regression line equation was developed to estimate the area affected by landslides during earthquakes in NZ: $\text{Log}_{10} A \text{ (area km}^2\text{)} = 0.96 M \text{ (magnitude)} - 3.7$. From this relationship the areas likely to be affected by landsliding for shallow earthquakes (depth $<45 \text{ km}$) of different

magnitude are: M 5, ~50-100 km²; M 6 - ~300-500 km²; M 7, ~1000-3000 km²; M 7.5-7.8, ~3000-7000 km²; M 8 or >, up to ~25,000 km² (see Figure 6).

This relationship differs slightly from that for overseas earthquakes (Keefer, 1984), which generally affect larger areas (Figure 5), probably because of the poorly understood interactions of topographic, geologic, and climatic factors.

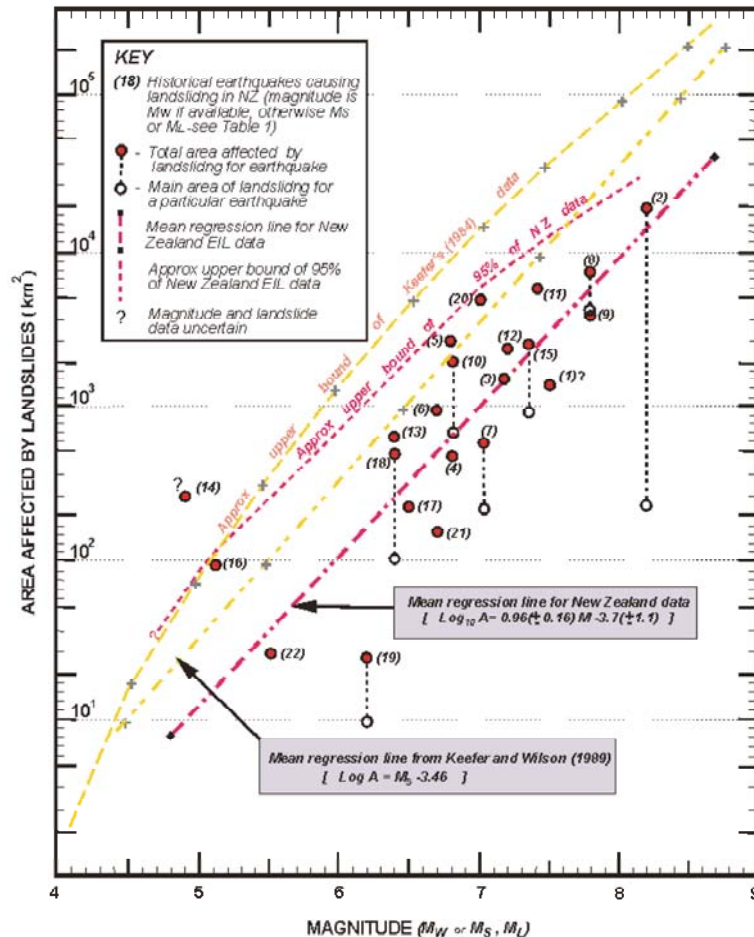


Figure 5: Relationship of the area affected by landslides during historical earthquakes of different magnitude in New Zealand (from Hancox et al., 2002).

Figure 6 shows the area (~25,000 km²) affected by landslides during the 1855 Wairarapa earthquake, the largest historical earthquake in New Zealand. It also shows the concentration of shallow debris slides in the epicentral area on steep slopes of the southern Rimutaka Range. These landslides are frequently reactivated by rain storms.

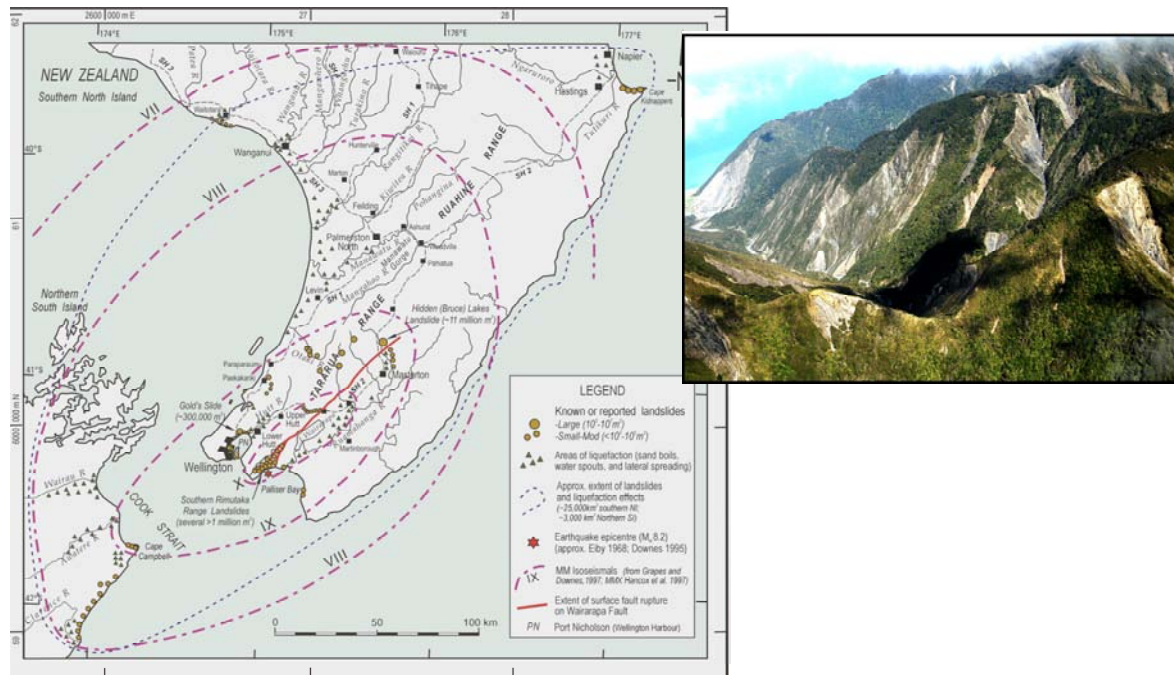


Figure 6: Map showing the known landslides and liquefaction effects caused by the 1855 Wairarapa earthquake (M 8.2). The low density of landslides is due to the sparse mid 19th century population. Landslides in the epicentral area of the southern Rimutaka Range are shown in the photo above. (after Hancox *et al.*, 1997)

2.1 LIQUEFACTION EFFECTS:

The intensity threshold for liquefaction in New Zealand (Figure 7) was found to be MM7 for sand boils (ejections of sand and water from a central point), and MM8 for lateral spreading (ground fissuring with sand and water ejections, extension, and subsidence). Both can occur at one intensity level lower in highly susceptible materials. The minimum magnitude for liquefaction is M 6, but is more common at M 7 and greater. Liquefaction ground damage is most common at MM8-10, at epicentral distances of 10-100 km.



Sand boil (MM8, Edgecumbe, 1987)



Minor lateral spreading (MM8, Edgecumbe, 1987)



Major lateral spreading (MM9, Napier, 1931)



Lateral spreading damage (MM10, Napier, 1931)

Figure 7: Examples of liquefaction effects during earthquakes in New Zealand. *Sand boils* (top left) are non-damaging and occur at MM7 and rarely at MM6. *Lateral spreading* occurs at MM7–8 and above. Major spreading damage occurs at MM9–10, as occurred during the M 7_w 7.8 Napier earthquake in 1931 (bottom).

2.2 LANDSLIDE SIZE

Landslide size is strongly dependent on magnitude, intensity, and distance, as well as slope angle and height. Large and very large landslides occur at distances of up to about 100 km (for M 6 or more, at MM8–10). Landslides during earthquakes overseas often occur at greater maximum distances, probably due to poorly understood interactions between topography, geology, climate, and seismic factors. Landsliding in New Zealand is often more widespread during winter, or after heavy rain. Rainfall before an earthquake can substantially increase the number and size of landslides caused by the earthquake (Dellow and Hancox, 2006).

After earthquake magnitude and shaking intensity, landslide occurrence is most strongly controlled by the topography, and rock and soil types. Most failures occur on moderate to very steep slopes (20°–50°) greater than 6–10 m high. The most common earthquake-induced landslides are rock and soil falls on steep cliffs, scarps, gravel banks and terrace edges, and high (>6 m) unsupported man-made cuts. Such features are highly hazardous and more susceptible to rapid failure because of rock defects, low strength, and topographic amplification of earthquake shaking. Large dip slope failures of Tertiary sandstone and mudstone often occur on gentle to steep slopes (10°–40°). Very large rock avalanches are caused by earthquakes of M 6.5 or greater, on slopes steeper than 25°–30° and more than 100–200 m high, especially on strongly shaken high narrow ridges. The occurrence of different types of landslides during earthquakes in NZ is summarised in Table 1.

Table 1: Occurrence and slope angles for typical earthquake-induced landslides New Zealand.

Landslide Type	Occurrence and characteristics (size, slope angle)
Rock / debris falls	Most common, very small to large. Minimum slope angle ~40°
Rock / debris slides	Very common, very small-large, minimum slope 25-30° (rare <20°).
Rock and debris avalanches	Very common, mod to very large, minimum slope 25°, more commonly 35-40° or >. Minimum slope height ~150 m.
Rotational slides	Moderately common. Minimum slope angle 15°.
Dip slope slides	Common in Tertiary sandstones and mudstones dipping ~25-35°.
Rock block slides	Uncommon, large. Minimum slope angle 5-10° generally.
Slow earth / mudflows	Uncommon. Minimum slope angle 10°.
Rapid soil flows	Uncommon in, but common overseas in susceptible materials.
Landslide size: Very small (< 10 ³ m ³); small (10 ³ –10 ⁴ m ³); Moderate (10 ⁴ m ³); Large (10 ⁵ m ³); Very large (10 ⁶ m ³ or >).	

Table 2: Slope angles at which landslides occur in New Zealand³

Slope Class	Rainfall-induced Landslides (2004 Southern Nth Island Rainstorm) ¹	Historical EILs in New Zealand ²
GENTLE (0 – 15°)	1.5 %	1 %
MODERATE (16 – 25°)	71 %	9 %
STEEP 26 – 35°	27 %	39 %
VERY STEEP (>35°)	0.5 %	60 % (36–45 ° = 40%; >45 ° = 20%)
1. Tertiary Hill Country (Hancox and Wright, 2005). 2. Hancox et al 1997, 2002. 3. Natural and man-made slopes.		

As shown in Tables 1 and 2, most landslides occur on moderate to steep slopes (~15°–>35°), including cliffs, scarps, terrace edges, and particularly road cuts and excavations. For some events and terrain there is a clear difference between rainfall and earthquake-induced landslide occurrence. More EILs occur on very steep rock slopes (cuts, scarps) and 25°–35° dip slopes.

A good correlation was found between landslide distribution and the fault rupture zone defined by aftershocks. This suggests that landslide distribution can be used to indicate the location of the earthquake epicentre and fault rupture zone, but allowance must be made for topographic and geologic effects (i.e. steep, landslide susceptible slopes or materials).

3 LANDSLIDING IN THE MM INTENSITY SCALE

The 1997 study of 22 historical earthquakes enabled environmental criteria in the New Zealand MM intensity scale to be defined in more detail. This was done by determining the characteristic landslide types and sizes in different MM zones established from building damage. For earlier earthquakes this allowed intensity maps based on landsliding to be prepared and intensities assigned in areas where there were few buildings (e.g., Murchison earthquake, Figure 2). Environmental criteria in the New Zealand MM scale are summarised in Table 3, and a complete list is presented in Appendix 1. These criteria have now been adopted in NZ (Dowrick et al., 2008), and accepted internationally in MM scales based on environmental effects (Guerrieri and Vittori, 2007). The terms used to describe landslide size (Tables 1 and 3) have also been adopted.

Table 3: Summary of Landsliding and Liquefaction effects in the New Zealand MM Intensity Scale.

MM Intensity	Main Environmental Response Criteria
MM 6	■ Isolated falls of boulders very small ($10^1 - < 10^3 \text{ m}^3$) rock falls/slides on cuts and banks. ■ Minor liquefaction effects - a few small sand boils in the most susceptible deposits.
MM 7	■ Small-moderate landslides ($10^3 - 10^5 \text{ m}^3$) & rock falls, especially on unsupported cuts. ■ Small sand boils are common in susceptible materials.
MM 8	■ Widespread smaller landslides; some moderate to very large failures ($10^5 - 10^6 \text{ m}^3$); ■ Significant failures on steep high cut slopes. Small landslide-dammed lakes formed. ■ Sand boils widespread; localised lateral spreading of river banks and lake shores.
MM 9 and 10 [MM10 similar to MM9, damage more severe]	■ Widespread damaging landslides, some large to extremely large ($10^5 - 10^7 \text{ m}^3$ or $>$), ■ Landslide-dammed lakes formed in susceptible terrain (possible dam-breach risk). ■ Sand boils widespread on alluvium, with lateral spreading (cracking, water and sand ejections, and slumping) along river banks, canals, and water-front areas.
Notes: Full descriptions of the criteria for MM6 to MM10 are presented in Appendix 1. Criteria for MM11 and MM12 are not defined as these intensities have not been reported in New Zealand (Dowrick et al., 2008).	

From the relationships derived from EILs in New Zealand, 5 Classes (*very low, low, moderate, high, and very high*) of EIL Opportunity have been recognised (Figure 8). These classes reflect the relative likelihood of landslides occurring at different MM intensities and equivalent peak ground acceleration (PGA) in New Zealand (based on the PGA/MM correlations of Murphy and O'Brien, 1977). This relationship allows earthquake-induced landslides to be used to roughly estimate the peak ground acceleration in an earthquake area. However, topographic amplification (on high ridges, steep slopes) and soft ground resonance effects also have to be considered.

The refined environmental response criteria have provided more certainty and consistency in assigning MM intensities from earthquake ground damage in areas where there are few buildings. They have recently been used to prepare isoseismal maps for the 2003 (M_w 7.2) and 2009 (M_w 7.8) earthquakes in Fiordland (Hancox et al., 2004; Reyners et al., 2003).

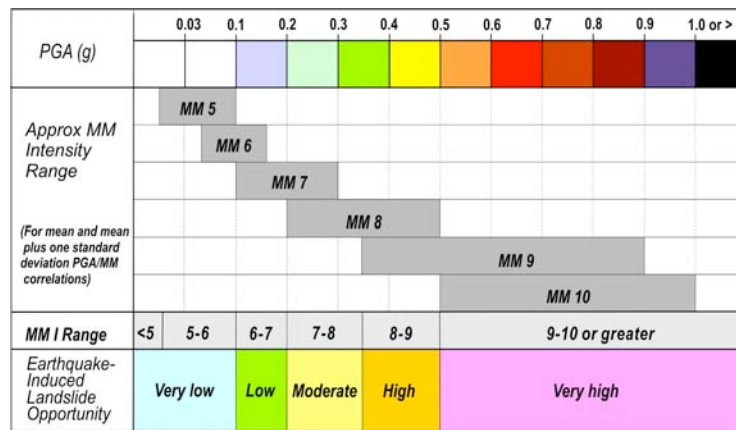


Figure 8: Correlation of PGA and MM Intensity has been used to define EIL Opportunity Classes in New Zealand.

4 SEISMICITY AND EARTHQUAKE-INDUCED LANDSLIDE POTENTIAL IN ADELAIDE

In the last 100 years South Australia has been shaken by 10 earthquakes greater than magnitude 5. The largest historical earthquake was the Beachport-Kingston earthquake of 1897 (M_L 6.5) ~300 km southeast of Adelaide (Greenhalgh and Singh, 1988), where it was felt only weakly. Most earthquakes felt in Adelaide have magnitudes less than M 3 and are barely perceptible (MM1–2). The epicentres of micro-earthquakes show a correlation with known faults, especially the Para Fault which runs through NW Adelaide. The ‘Dirt Scarp’ of the Para Fault suggests it may have been active in the last 10,000 years, and future seismic activity along it is possible (Selby and Lindsay, 1981).

The last big earthquake near Adelaide was of magnitude 5.5, on 1 March 1954. It caused widespread minor damage to buildings, with a cost of \$8 million (Sinadinovski, *et al.*, 2006, Mc Hue, 1975) or possibly ~\$80 million today. The earthquake is believed to have been centred on the Eden-Burnside Fault at Darlington, about 5 km S of Adelaide CBD, where the strongest shaking was recorded (Kerr-Grant, 1956; Malpas, 1991; Sinadinovski *et al.*, 2006).

The 1954 earthquake caused building damage consistent with maximum MM 8 intensity shaking (fallen chimneys, parapets cracked masonry, partial wall collapses) over a small area near the epicentre. A few small soil slides and cracks (~100 m long, 1-2 cm wide) were noted on steep slopes of the Eden-Burnside Fault scarp (MM 7-8 zone). Very minor liquefaction effects (sand and water ejections, new springs) and small river bank slides were reported at Mt Barker Creek (MM 6-7 zone), ~15 km east of the epicentre (Sinadinovski *et al.*, 2006). There were no reports of any moderate or large landslides, or failures of road cuts caused by the earthquake. However it is likely that some road cuts were affected by the earthquake, especially in Adelaide Hills, but these small failures have not been reported.

Future larger earthquakes (perhaps ~ M 6.0–7.0) may be associated with the Para Fault or other faults in the area, resulting in shaking intensities of up to MM 8–9 across the Adelaide area. Although the historical seismicity of the region suggests the probability of magnitude 6–7 earthquakes in Adelaide is low, if such an earthquake does occur, with a maximum intensity of MM8–9, the possible landsliding and liquefaction effects could be significant. There could, for example, be moderate to large rock falls and slides on high unsupported road cuts, such as those on the Princess Highway southeast of Adelaide (Figure 9). Smaller failures are also likely at MM 7 (cuts probably not present in 1954). Some cuts were rock-bolted (pers. comm. Dave Stapledon, 2009), and these will be less prone to failure during earthquakes.



Figure 9: High unsupported cut slopes along the Princess Highway southeast of Adelaide are likely to suffer moderate to large rock falls under MM 8-9 intensity shaking (*left photo by Tony Meyers*).

High, unsupported cuts in rock quarries in the hills east of Adelaide could also be affected by slope failures during strong earthquake shaking. The quarry shown in Figure 10 is in competent dolomite and, although large landslides are probably not likely at this site, block falls and small overburden failures are possible.



Figure 10: Cut slopes in quarries in the hills east of Adelaide could fail during very strong (MM8-9) earthquake shaking (*Google Earth image*).

Moderate to large shallow rock falls and slides could also at occur Sellick's Hill on the main road south of Adelaide, on both natural slopes and high, steep cut slope (Figures 11 and 12). There could also be failures on river banks, terraces edges, coastal cliffs, and unsupported excavations for buildings across the strongly shaken region.



Figure 11: Google Earth image showing steep natural slopes and road cuts at Sellick's Hill, on the main road south of Adelaide. Both natural slopes and unsupported cut slopes shown here are potentially vulnerable to large failures during strong earthquakes (*Google Earth image*).

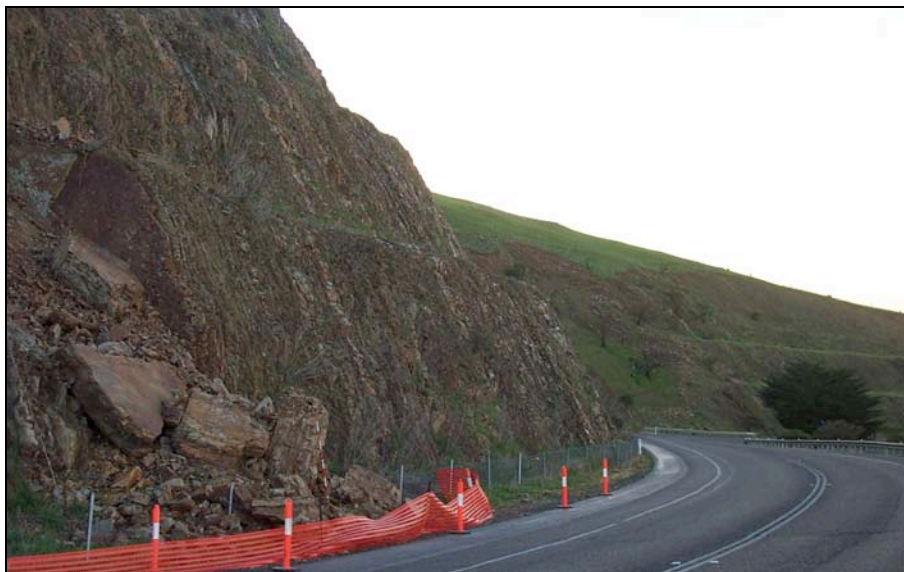


Figure 12: High, steep cuts on the Main South Road at Sellicks Hill south of Adelaide could be significantly damaged by moderate to large rock falls if subjected to MM 8-9 shaking. Failures are also likely on similar high and unsupported road and rail cuts throughout the Adelaide area (*photo by Tony Meyers*).

Significant liquefaction (lateral spreading and subsidence) damage could also occur in the region, especially in areas underlain by saturated sandy alluvium, and low lying estuarine deposits, such as along the banks of the Torrens and Para rivers, and along the western coast line. The Port of Adelaide and Adelaide Airport may also be vulnerable to earthquake damage if the underlying deposits have high liquefaction susceptibility (Figure 13).



Figure 13: Liquefaction susceptibility of sandy alluvium and low lying estuarine deposits in the Adelaide area may be potentially high, especially in areas adjacent to the Torrens River, coastal areas, including Port Adelaide, and Adelaide Airport (L) (Google Earth image).

There is moderate seismic hazard potential in the Adelaide area. Earthquake-induced landslide and liquefaction susceptibility studies may therefore be required to locate high hazard areas, if such studies have not already been undertaken. The historical seismic record, however, suggests that the probability of M 6-7 earthquakes and potential risk from earthquake-induced landslides and liquefaction in Adelaide is likely to be low.

5 SUMMARY

Studies of landsliding and soil liquefaction caused by historical earthquakes in New Zealand have shown that the minimum magnitude for earthquake-induced landsliding (EIL) in NZ is M 5, with significant landsliding occurring at M 6 or greater. The minimum intensity for landsliding is MM6, and the most common intensities MM 7–8. The threshold for liquefaction is M 6 and MM 7 for sand boils, and MM 8 for lateral spreading, but both can occur at an intensity level lower in highly susceptible materials. Areas likely to be affected by shallow (<45 km) earthquakes of different magnitude are: M 5, ~50–100 km²; M 6 - ~300–500 km²; M 7, ~1000–3000 km²; M 7.5–7.8, ~3000–7000 km²; M 8 or >, up to ~25,000 km².

Environmental response criteria (landsliding, subsidence, liquefaction, lateral spreading) have been defined for the New Zealand MM Intensity Scale. These are now widely used to assign MM intensities in earthquake areas where there are few buildings, or to estimate the likely severity of landsliding at different levels of earthquake shaking. Landslide size is strongly dependent on magnitude, intensity and distance, as well as slope angle and height. Large and very large landslides occur at distances of up to about 100 km (for M 6 or more, at MM 8–10). Most failures occur on moderate to very steep slopes (20°–50°) greater than 6–10 m high. The most common earthquake-induced landslides are rock and soil falls on steep cliffs, scarps, gravel banks and terrace edges, and high (>6 m) unsupported man-made cuts.

The EIL relationships developed in New Zealand suggest that an M 6–7 (intensity MM 8–9) earthquake in the Adelaide area could cause moderate to large rock falls and slides on high unsupported road cuts in the area, and failures of river banks, terrace edges, coastal cliffs, and unsupported excavations. Significant liquefaction (lateral spreading and subsidence) damage could also occur, especially in areas underlain by saturated sandy alluvium, and low lying estuarine deposits. However, the historical seismic record suggests that the probability of an M 6 or M 7 earthquake in Adelaide is relatively low; hence the potential risk from earthquake-induced landslides and liquefaction is also likely to be low.

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APPENDIX 1

Landslide and Environmental Criteria for the Modified Mercalli Intensity Scale – NZ 2007	
MM5	Loose boulders may occasionally be dislodged from steep slopes.
MM6	Trees and bushes shake, or are heard to rustle. Loose material may be dislodged from sloping ground, e.g. existing slides, talus and scree slopes.

	A few very small ($\leq 10^3$ m³) soil and regolith slides and rock falls from steep banks and cuts. A few minor cases of liquefaction (sand boil) in highly susceptible alluvial and estuarine deposits.
MM7	Water made turbid by stirred up mud. Small slides such as falls of sand and gravel banks, and small rock-falls from steep slopes and cuttings common. Instances of settlement of unconsolidated, or wet, or weak soils. A few instances of liquefaction (ie. small water and sand ejections). Very small ($\leq 10^3$ m³) disrupted soil slides and falls of sand and gravel banks, and small rock falls from steep slopes and cuttings are common. Fine cracking on some slopes and ridge crests. A few small to moderate landslides (10^3 – 10^5 m³), mainly rock falls on steeper slopes ($>30^\circ$) such as gorges, coastal cliffs, road cuts and excavations. Small discontinuous areas of minor shallow sliding and mobilisation of scree slopes in places. Minor to widespread small failures in road cuts in more susceptible materials. A few instances of non-damaging liquefaction (small water and sand ejections) in alluvium.
MM8	Cracks appear on steep slopes and in wet ground. Significant landsliding likely in susceptible areas. Small to moderate (10^3-10^5 m³) slides widespread; many rock and disrupted soil falls on steeper slopes (steep banks, terrace edges, gorges, cliffs, cuts etc). Significant areas of shallow regolith landsliding, and some reactivation of scree slopes. A few large (10^5-10^6 m³) landslides from coastal cliffs, and possibly large to very large ($\geq 10^6$ m³) rock slides and avalanches from steep mountain slopes. Larger landslides in narrow valleys may form small temporary landslide-dammed lakes. Roads damaged and blocked by small to moderate failures of cuts and slumping of road-edge fills. Evidence of soil liquefaction common, with small sand boils and water ejections in alluvium, and localised lateral spreading (fissuring, sand and water ejections) and settlements along banks of rivers, lakes, and canals etc. Increased instances of settlement of unconsolidated, or wet, or weak soils.
MM9	Cracking of ground conspicuous. Landsliding widespread and damaging in susceptible terrain, particularly on slopes steeper than 20°. Extensive areas of shallow regolith failures and many rock falls and disrupted rock and soil slides on moderate and steep slopes (20°-35° or greater), cliffs, escarpments, gorges, and man-made cuts. Many small to large (10^3-10^6 m³) failures of regolith and bedrock, and some very large landslides (10^6 m³ or greater) on steep susceptible slopes. Very large failures on coastal cliffs and low-angle bedding planes in Tertiary rocks. Large rock/debris avalanches on steep mountain slopes in well-jointed greywacke and granitic rocks. Landslide-dammed lakes formed by large landslides in narrow valleys. Damage to road and rail infrastructure widespread with moderate to large failures of road cuts and slumping of road-edge fills. Small to large cut slope failures and rock falls in open mines and quarries. Liquefaction effects widespread with numerous sand boils and water ejections on alluvial plains, and extensive, potentially damaging lateral spreading (fissuring and sand ejections) along banks of rivers, lakes, canals etc). Spreading and settlements of river stop-banks likely.
MM10	Landsliding very widespread in susceptible terrain. Similar effects to MM9, but more intensive and severe, with very large rock masses displaced on steep mountain slopes and coastal cliffs. Landslide-dammed lakes formed. Many moderate to large failures of road and rail cuts and slumping of road-edge fills and embankments may cause great damage and closure of roads and railway lines. Liquefaction effects (as for MM9) widespread and severe. Lateral spreading and slumping may cause rents over large areas, causing extensive damage, particularly along river banks, and affecting bridges, wharfs, port facilities, and road and rail embankments on swampy, alluvial or estuarine areas.
Notes: (1) "Some or 'a few' indicates that threshold for response has just been reached at that intensity. (2) Environmental damage (response criteria) occurs mainly on susceptible slopes and in certain materials, hence the effects described above may not occur in all places, but can be used to reflect the average or predominant level of damage or MM intensity in an area. (3) Environmental criteria not defined for MM11 and 12, as those intensities have not been reported in New Zealand. Earlier versions of the MM intensity scale suggest that environmental effects at MM11-12 are similar to MM9- 10, but are possibly more widespread and severe. (4) This appendix is based on Hancox et al. 1997, 2002, and Dowrick et al., 2008.	